

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report evaluates patent evidence relevant to low acrylonitrile nitrile butadiene rubber (NBR), with emphasis on composition control, monomer feed strategy, and emulsion polymerization process design. The target material is a butadiene-acrylonitrile copolymer with reduced acrylonitrile fraction relative to conventional NBR grades, typically achieved by controlling acrylonitrile feed, conversion profile, and reactor staging. In the provided manifest, only one primary patent was identified: US10208151B2. Although the patent is not directed to standalone low-ACN NBR, it is highly relevant because it discloses a diene-based rubber latex prepared by staged emulsion polymerization of 1,3-butadiene with controlled addition of emulsifier, initiator, electrolyte, molecular weight regulator, and cross-linking agent, followed by grafting with acrylonitrile/styrene to form ABS. The disclosed process demonstrates how rubber-phase latex properties can be tuned through monomer charge, cross-linker type, addition timing, temperature staging, and conversion endpoints. Key process trends include two-step polymerization, nitrogen-substituted reactor operation, use of potassium rosinate/potassium oleate emulsification, potassium persulfate initiation, tertiary dodecyl mercaptan molecular-weight control, and termination at high conversion (90%+). The examples show that changing the cross-linking agent from propylene glycol diacrylate $n=7$ to $n=13$ and adjusting addition timing changes gel content and particle size while preserving large latex particle size (~ 3100 Å). For low-ACN NBR synthesis, the patent is useful as a process-control reference for emulsion copolymerization and latex property management, even though it does not explicitly disclose low acrylonitrile NBR compositions.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10208151B2
- Patent Title: Method for preparing large-size diene-based rubber latex and acrylonitrile- ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a two-stage emulsion polymerization route to prepare large-size diene-based rubber latex from 1,3-butadiene, using staged monomer/emulsifier addition, potassium persulfate initiation, tertiary dodecyl mercaptan molecular-weight control, and a cross-linking agent of (propylene glycol)_n diacrylate. The resulting latex is then used as the rubber phase for grafting acrylonitrile and styrene to make ABS copolymer powder and thermoplastic resin. - Ion exchanged water: 65 parts by weight — Example 1 - 1,3-butadiene: 75 parts by weight — Example 1 - Potassium rosinate: 1 part by weight — Example 1 - Potassium oleate: 1.5 parts by weight — Example 1 - Potassium carbonate (K₂CO₃): 0.3 part by weight — Example 1 - Tertiary dodecyl mercaptan (TDDM): 0.3 part by weight — Example 1 - Potassium persulfate (K₂S₂O₈): 0.3 part by weight — Example 1 - (Propylene glycol)₇ diacrylate: 0.2 part by weight — Example 1 - Reaction temperature: 70° C — Example 1 - First polymerization conversion rate: 40% — Example 1 - Reaction temperature: 82° C — Example 1 - Second polymerization conversion rate: 75% — Example 1 - Final polymerization conversion rate: 93% — Example 1 - Gel content: 78% — Example 1 - Particle size: 3100 Å — Example 1 - Diene-based rubber polymer: 60 parts by weight — Example 1 - Ion exchanged water: 100 parts by weight — Example 1 - Acrylonitrile: 10 parts by weight — Example 1 - Styrene: 30 parts by weight — Example 1 - Ion exchanged water: 25 parts by weight — Example 1 - t-Butyl hydroperoxide: 0.12 part by weight — Example 1 - Potassium rosinate: 1.0 part by weight — Example 1 - Tertiary dodecyl mercaptan (TDDM): 0.3 part by weight — Example 1 - Dextrose: 0.054 part by weight — Example 1 - Sodium pyrophosphate: 0.004 part by weight — Example 1 - Ferrous sulfate: 0.002 part by weight — Example 1 - Reaction temperature: 70° C — Example 1 (ABS grafting) - Addition time: 3 hours — Example 1 - Dextrose: 0.05 part by weight — Example 1 - Sodium pyrophosphate: 0.03 part by weight — Example 1 - Ferrous sulfate: 0.001 part by weight — Example 1 - t-Butyl hydroperoxide: 0.05 part by weight — Example 1 - Reaction temperature: 80° C — Example 1 - Polymerization conversion rate: 97% — Example 1 - Graft rate: 41% — Example 1 - Coagulates content: 0.3% — Example 1 - (Propylene glycol)₁₃ diacrylate: 0.2 part by weight — Example 2 - Gel content: 76% — Example 2 - Particle size: 3100 Å — Example 2 - (Propylene glycol)₁₃ diacrylate: 0.2 part by weight — Example 3 - (Propylene glycol)₁₃ diacrylate: 0.1 part by weight — Example 3 - Reaction temperature: 70° C — Example 3 - First polymerization conversion rate: 40% — Example 3 - Reaction temperature: 82° C — Example 3 - Final polymerization conversion rate: 93% — Example 3 - Gel content: 70% — Example 3 - Particle size: 3100 Å — Example 3 - Diene-based rubber latex: 40 wt% to 70 wt% — Claims - Aromatic vinyl compound: 20 wt% to 50 wt% — Claims - Vinyl cyanide compound: 10 wt% to 40 wt% — Claims - Conjugated diene-based monomer: 50 parts by weight to 75 parts by weight — Claims - Emulsifier: 1 part by weight to 3 parts by weight — Claims - Polymerization initiator: 0.1 part by weight to 0.4 part by weight — Claims - Electrolyte: 0.1 part by weight to 3 parts by weight — Claims - Molecular weight control agent: 0.1 part by weight to 0.5 part by weight — Claims - Ion exchanged water: 65 parts by weight to 100 parts by weight — Claims - Cross-linking agent: 0.05 part by weight to 0.3 part by weight — Claims - Balance monomer addition: 0 to 0.1 parts by weight emulsifier — Claims - Balance cross-linking agent: 0 to 0.2 parts by weight — Claims - Step 1 conversion point: 45% to 60% — Description - Step 3 termination conversion: 90% to 95% or

higher — Description - Average particle diameter: 2,500 Å to 4,000 Å — Description - Gel content: 60% to 80% — Description

Relevant Experimental Evidence - Example 1: Produced large-size rubber latex with 78% gel content and 3100 Å particle size using (propylene glycol)7 diacrylate, then grafted acrylonitrile/styrene to ABS latex with 97% conversion and 41% graft rate. - Example 2: Replacing the cross-linker with (propylene glycol)13 diacrylate lowered gel content to 76% while maintaining 3100 Å particle size. - Example 3: Using (propylene glycol)13 diacrylate in a split-addition mode further reduced gel content to 70% while preserving 3100 Å particle size. - Comparative Example 1: A non-crosslinked latex was prepared for comparison and then used in ABS grafting. - Comparative Example 2: Adding polypropylene glycol diacrylate (Mn=6,000) at 0.2 part by weight produced a large-size latex of 3100 Å for comparison.

Technical Relevance This patent is relevant because it provides a detailed emulsion-polymerization framework for controlling diene rubber latex morphology, conversion, and grafting behavior, all of which are important when designing low-ACN NBR or related nitrile rubber systems. While it does not explicitly claim low acrylonitrile NBR, its staged addition strategy and composition control principles are directly transferable to acrylonitrile/butadiene copolymer synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence base contains one primary patent, US10208151B2, which is not a direct low-acrylonitrile NBR disclosure but is technically useful for nitrile-rubber process design. The strongest transferable lessons are staged emulsion polymerization, controlled monomer and emulsifier addition, use of a molecular-weight regulator, and conversion-based termination to manage latex size and gel content. For low-ACN NBR synthesis, the most relevant evidence-supported process direction is careful feed-ratio control in emulsion polymerization, with staged addition and temperature management to maintain composition uniformity and polymer properties. However, no patent in the provided manifest explicitly discloses a low-acrylonitrile NBR composition, so any low-ACN target specification must be inferred from broader nitrile-rubber process principles rather than directly claimed composition data.

5. REFERENCES

- US10208151B2 | Method for preparing large-size diene-based rubber latex and acrylonitrile- ...
| Not disclosed | US | 2019 | <https://patents.google.com/patent/US10208151B2/en>